Unit 5: Trigonometry

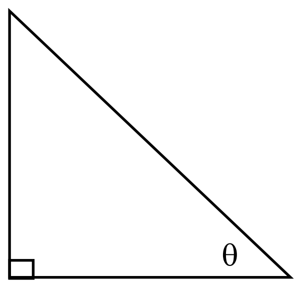
Homework Log

Lesson	Date	Section	Assigned Work
1		Review Primary Trig Ratios	LP: pg 4 #1 – 9
2		The Sine Law	Text: pg 318 #1a, 4a,7, 8 LP: pg 7 #1 – 6
3		The Cosine Law	Text: pg 325 #1 – 10, 14
4		Secondary Trigonometric Ratios	Text: pg 280 #1 – 6, 9, 10, 18 – 20
5		The Ambiguous Case of the Sine Law	Text: pg 318 #3, 5
6		Exploring Trigonometric Ratios for Angles Between 0° than 360°	Text: pg 299 #1 – 6, 9 – 15
7		Exploring Trigonometric Ratios for Angles Greater Than 360°	LP: pg 20 #1 – 8
8		Special Angles: Exact Trigonometric Ratios	LP: pg 26 #1 – 10 Text: pg 287 #11 – 14
9		Review	LP: pg 28 #1-15 Text: pg 339 #8 – 13 pg 340 #5 – 8

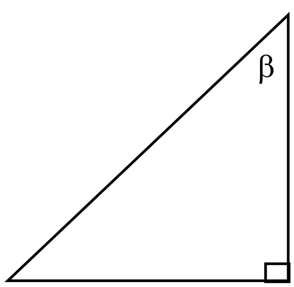
Homework tip: Don't ask.

Trigonometric Ratios

Are for



1. Label the sides of the triangle as__________, ___________ or _____________ for the given/indicated angle.



2. Choose the _____ that included the two sides you have labelled.
3. Solve for the _________ side or angle.

A mnemonic device to help remember the trig ratios:

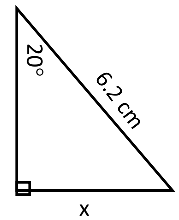
SOH CAH TOA

Trig ratios only work for ______ triangles!

Sine

$\sin(\theta) =$

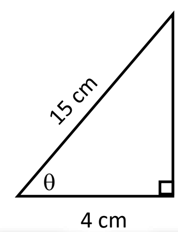
$\theta = \sin^{-1}(\quad)$



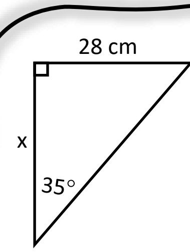
Cosine

$\cos(\theta) =$

$\theta = \cos^{-1}(\quad)$

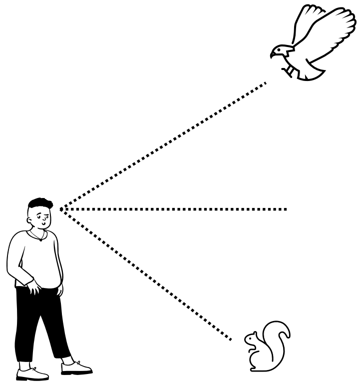


Tangent



$\tan(\theta) =$

$\theta = \tan^{-1}(\quad)$



Applications of Trigonometry

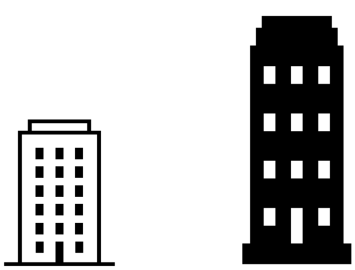
SOH CAH TOA

1. Draw a diagram if not given.
2. Label the diagram.
3. Choose the appropriate trig ratio.
4. Solve.

A 15 m long guy wire is attached to the top of a cell phone tower that is 11.5 m tall. Determine the angle made with the wire and the ground.

Tayari is ziplines down a zipline with an angle of depression of 8.5°. If the line is 500 ft long, how far will Tayari travel horizontally?

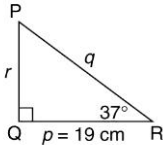
Akela stands on the roof of the shorter of two buildings that are 75 m apart. They measure the angle of elevation to the taller building to be 46° and the angle of depression the base of the other building to be 32°. Determine the heights of both buildings.



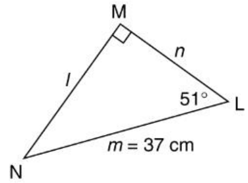
Reviewing Right Triangles

1. Solve each triangle. Round lengths to the nearest unit and angles to the nearest degree.

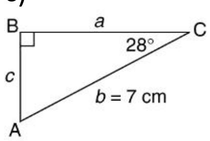
a)



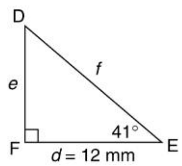
b)



c)

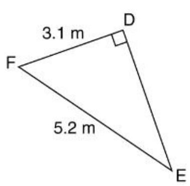


d)

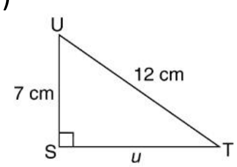


2. Solve each triangle. Round lengths to the nearest unit and angles to the nearest degree.

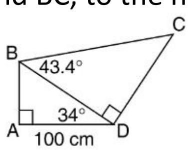
a)



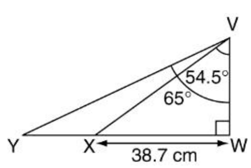
b)



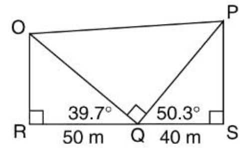
3. Find BC, to the nearest centimetre.



4. Find XY, to the nearest tenth of a centimetre



5. Find PO, to the nearest tenth of a metre.



6. To guard against falls, an extension ladder should make an angle of 75° or less with the ground. What is the maximum height that an 8-m ladder can reach safely?

7. A patient is being treated with laser therapy to remove a tumour that is located behind a vital organ. The tumour is 5.7 cm below the skin. To avoid the organ, the entry point of the laser must be 12.4 cm above the tumour along the skin. What is the angle the doctor must use to hit the tumour directly?

8. A jet airplane begins a steady climb at 17° and flies for 5 km, measured along the ground. What is its change in altitude?

9. The city is designing a new multi-level parking garage. The floors are to be 4 m apart. The connecting ramps will be 22 m long. What is the angle of elevation for each ramp?

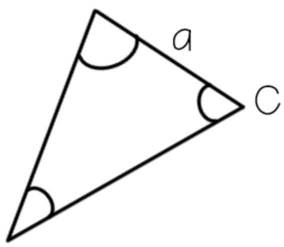
Answers

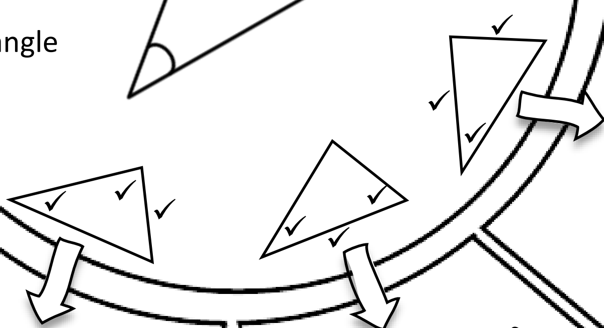
1. a) ∠P = 53°, *r* = 14 cm, *q* = 24cm b) ∠N = 39°, *l* = 29cm, *n* = 23cm c) ∠A = 62°, *a* = 6cm, *c* = 3cm d) ∠D = 49°, *e* = 10mm, *f* = 16mm
2. a) *f* = 4 *m*, ∠E = 37°, ∠F = 53° b) *u*= 10 cm, ∠U = 54°, ∠T = 36°
3. 166 cm
4. 20.5 cm
5. 90.2 m
6. 7.7 m
7. 25°
8. 1.5 km
9. 10.5°

Law of Sines

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

finish labeling the triangle

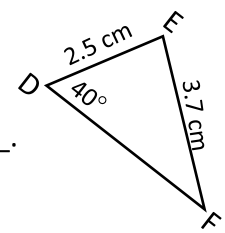




The sum of the 3 angles of a triangle = ____.
The sum of two sides of a triangle must always be _____________________ ________________________.

SSA

Solve the triangle



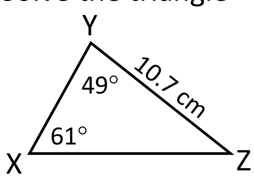
Step 1: Find angle __.

Step 2: Find angle __.

Step 3: Find side __.

AAS

Solve the triangle



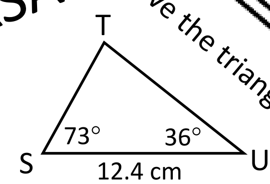
Step 1: Find side __.

Step 2: Find angle __.

Step 3: Find side __.

ASA

Solve the triangle



Step 1: Find angle __.

Step 2: Find side __.

Step 3: Find side __.

A pet's collar contains a transmitter that is helping its owners find the lost animal. From positions 250 m apart, the searchers can detect the transmissions at angles of 87° and 53° from the straight line between the searchers. How far is each searcher from the lost pet?

Draw a diagram.

Solve

A cellular relay tower 12.4 m high is placed on top of a building. Standing at a point some distance away from the building Jordie measures the angle of elevation to the top of the tower to be 48.2° and the angle of elevation to the bottom of the tower to be 42.5°. What is the height of the building?

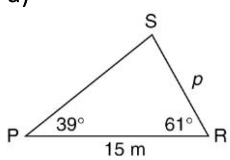
Draw a diagram.

Solve

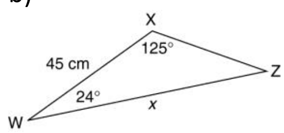
Sine Law Practice

1. Find the length of each indicated side, to the nearest tenth.

a)

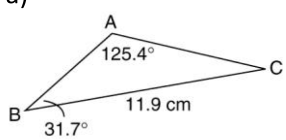


b)

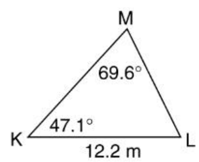


2. Solve each triangle. Round each calculated value to the nearest tenth of a unit, if necessary.

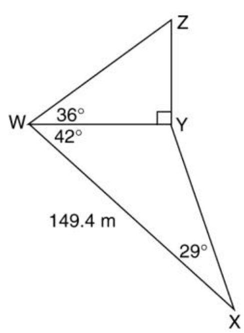
a)



b)



3. Surveying A surveyor collected the data in the diagram to find the height, *w*, of a monument. Find this height, to the nearest tenth of a metre.



4. Two radar stations that are 150 km apart located an unidentified plane that vanished from their screens at the same time. The first station indicated that the position of the plane made an angle of 44° with the line joining the stations. The second station indicated that the plane made an angle of 56° with the same line. How far is each station from the point where the plane disappeared, to the nearest kilometre?

5. A green house is 24 m wide. One side of the roof makes an angle of 25° with the horizontal, while the other makes an angle of 70°. Find the two rafter lengths.

6. From a window of a high-rise building, the angle of depression of a parked car is 24°. From a window 15 m lower, the angle of depression is 19°. How far from the foot of the building is the parked car?

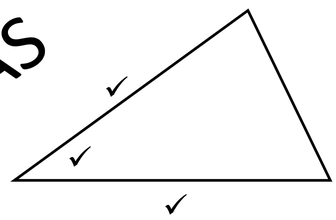
Answers

1. a) 9.6 m b) 71.6 cm 2. a) *b* = 7.7 cm, *c* = 5.7 cm, ∠C = 22.9°
b) *l* = 11.6 m, *k* = 9.5 m, ∠L = 63.3° 3. 55.7 m 4. 126 km, 106 km 5. 10 m, 23 m 6. 149 m

Law of Cosines

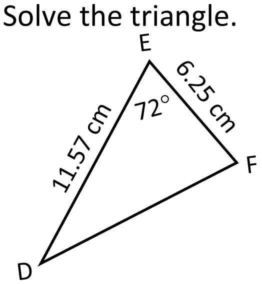
$$c^2 = a^2 + b^2 - 2ab\cos C$$

SAS



Contained Angle

Solve the triangle.

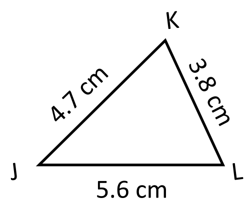


Step 1: Find side __.

Step 2: Find angle __.

Step 3: Find angle __.

Solve the triangle.

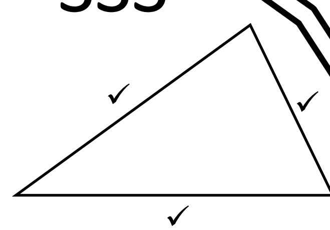


Step 1: Find angle __.

Step 2: Find angle __.

Step 3: Find angle __.

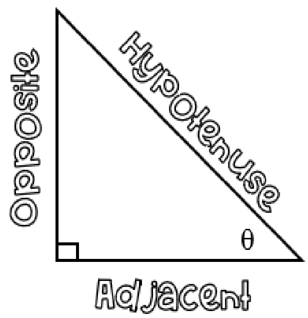
SSS



$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

A squash player hits the ball 2.3 m to the side wall. The ball rebounds at an angle of 100° and travels 3.1 m to the front wall. How far is the ball from the player when it hits the front wall? Assume the player does not move after her shot.

A piece of rope 19.5 m long is stretched taut to form a triangle with two sides 5.5 m and 7.3 m in length. Calculate the measure of the angle between these two sides of the triangle.



Secondary Trigonometric Ratios

The Primary Trigonometric Ratios

Sine

$$\sin(\theta) = \frac{O}{H}$$

Cosine

$$\cos(\theta) = \frac{A}{H}$$

Tangent

$$\tan(\theta) = \frac{O}{A}$$

The secondary trigonometric ratios are the ________________ of the primary trigonometric ratios.

Cosecant

$$\csc(\theta) = \frac{H}{O}$$

$$\csc(\theta) = \frac{1}{\sin(\theta)}$$

Secant

$$\sec(\theta) = \frac{H}{A}$$

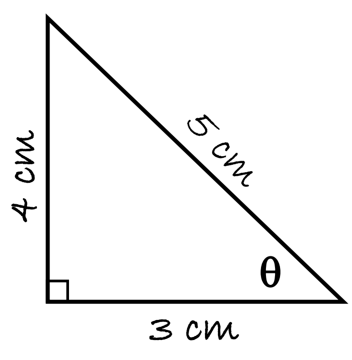
$$\sec(\theta) = \frac{1}{\cos(\theta)}$$

Cotangent

$$\cot(\theta) = \frac{A}{O}$$

$$\cot(\theta) = \frac{1}{\tan(\theta)}$$

State the six trigonometric ratios.



If $\cot A = \frac{5}{2}$, determine the remaining five trigonometric ratios.

Use a calculator to find the values of each of the following, correct to four decimal places.

a) $\cos 32°$ b) $\csc 57°$ c) $\sec 28°$ d) $\cot 15°$

Find the measure of the angle to the nearest degree, where $0° < \theta < 90°$.

a) $\sin\theta = 0.6215$ b) $\csc\theta = 1.8426$ c) $\sec\theta = \frac{13}{8}$ d) $\cot\theta = \frac{16}{25}$

The Ambiguous Case of the Sine Law

- **When given SSS, SAS and AAS these triangles are unique. There is only one way to construct these triangles.**
- **The triangle SSA may not be unique. It is possible that no triangle, 1 triangle, or 2 triangles exist with the given measurements. This situation is called the ambiguous case because more than one triangle can be drawn.**

When we have triangles that are SSA, how do we know how many triangles are possible?

A. When the opposite angle is obtuse, there are 2 cases to consider.

A1. If $a \leq c$

Construct $\triangle ABC$, where $\angle A = 120°$, $a = 3$ cm and $c = 5$ cm.

If $a \leq c$, then side a is ____________ to reach side b and thus, ______ triangle exists.

A2. If $a > c$

Construct $\triangle ABC$, where $\angle A = 120°$, $a = 8$ cm and $c = 5$ cm.

If $a > c$, then side a is _________ to reach side b at ____ place and thus, ___________ triangle exists.

B. When the opposite angle is acute, there are 4 cases to consider.

B1. If $a \geq c$

Construct $\triangle ABC$, where $\angle A = 30°$, $a = 7$ cm and $c = 6$ cm.

If $a \geq c$, then side a is _________ to reach side b at ____ place and thus, ___________ triangle exists.

B2. If $a = h$

Construct $\triangle ABC$, where $\angle A = 30°$, $a = 3$ cm and $c = 6$ cm.

If $a = h$, then side a is ___________________ to side b and thus, ______ triangle exists.

B3. If $a < h$

Construct $\triangle ABC$, where $\angle A = 30°$, $a = 2$ cm and $c = 6$ cm.

If $a \leq c$, then side a is ____________ to reach side b and thus, ______ triangle exists.

B4. If $h < a < c$

Construct $\triangle ABC$, where $\angle A = 30°$, $a = 4$ cm and $c = 6$ cm.

If $a \geq c$, then side a is _________ to reach side b at ____ place and thus, ___________ triangle exists.

SUMMARY

When solving a triangle given SSA, the number of solutions depends on the length of the side opposite the given angle and how it compares to the vertical height.

Case 1: Given angle is obtuse

i) $a > c$, then 1 triangle is possible

ii) $a \leq c$, then no triangle is possible

Case 2: Given angle is acute

i) $a \geq c$, then 1 triangle is possible

ii) If $a < c$, find the vertical height and compare it to the side opposite to the given angle.

- $a < h$, then no triangle is possible
- $a = h$, then 1 triangle is possible
- $a > h$, then 2 triangles are possible

Ex. 1: For each triangle, a set of measurements is given. Determine if there are zero, one, or two possibilities. Draw the triangle(s) to support your answer.

a) $\triangle ABC$: $\angle A = 55°$, $a = 7$ m, and $c = 12$ m

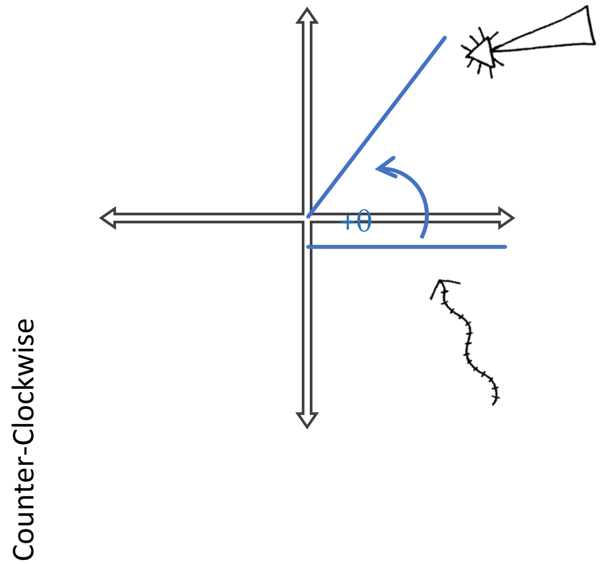
b) $\triangle DEF$: $\angle D = 50°$, $d = 10$ m, and $e = 6$ m

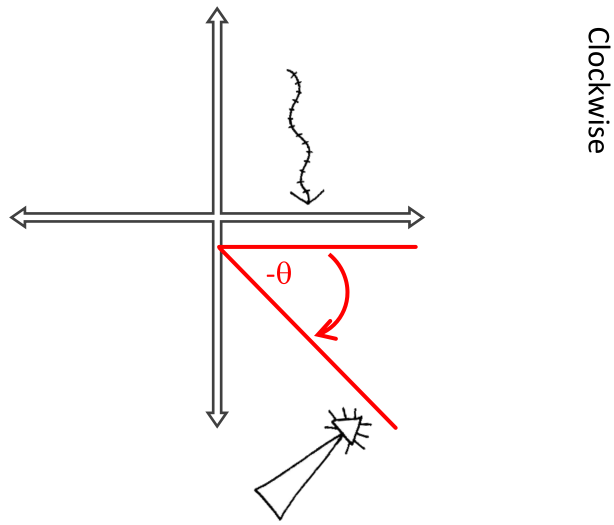
c) $\triangle KLM$: $\angle L = 115°$, $l = 3$ m, and $m = 9$ m

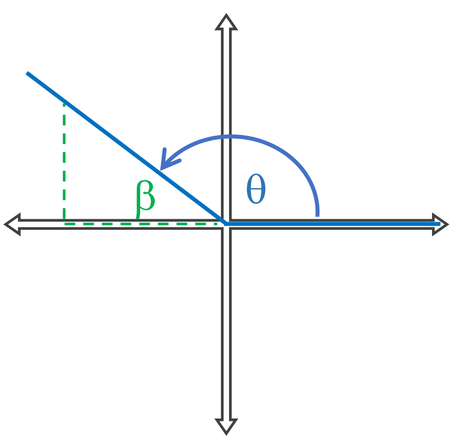
d) $\triangle STU$: $\angle U = 42°$, $u = 4$ m, and $t = 5$ m

Ex. 2: In $\triangle DEF$, $\angle D = 37°$, $f = 11$ cm, and $d = 7.5$ cm. Determine $\angle F$.

Exploring Trigonometric Ratios for Angles $0° \leq \theta \leq 360°$





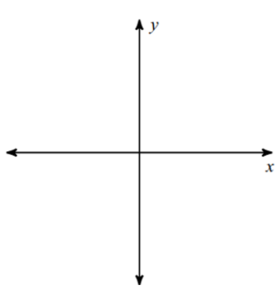


θ : referred to as the ____________ ________ in standard position. It is the angle between the initial arm (positive x-axis) and the terminal arm.

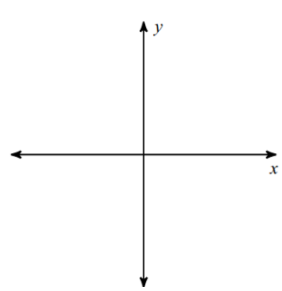
β : referred to as the ____________ ________ ________ (RAA) or reference angle. It is the angle between the x-axis and the terminal arm.

Sketch θ and determine β.

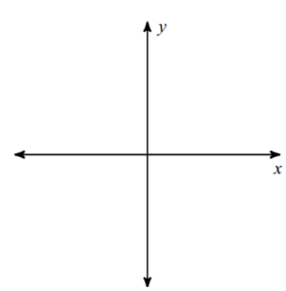
a) $\theta = 110°$



b) $\theta = 305°$

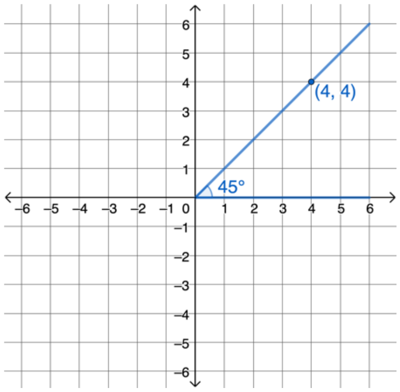


c) $\theta = -197°$

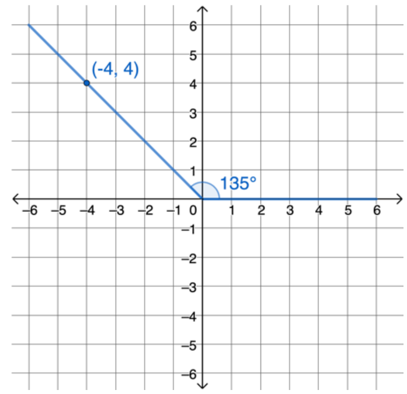


Let's investigate the trig ratios for each quadrant.

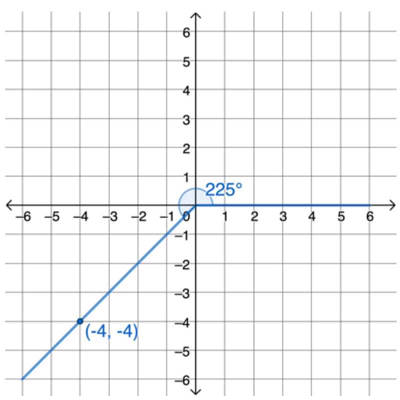
When θ is in Quadrant I.



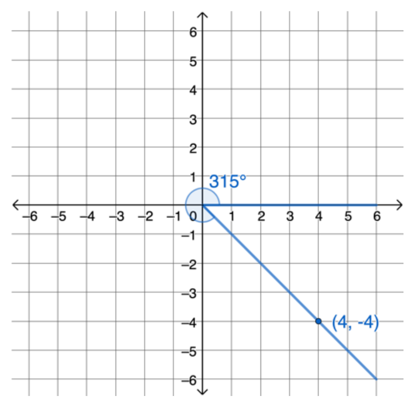
When θ is in Quadrant II.



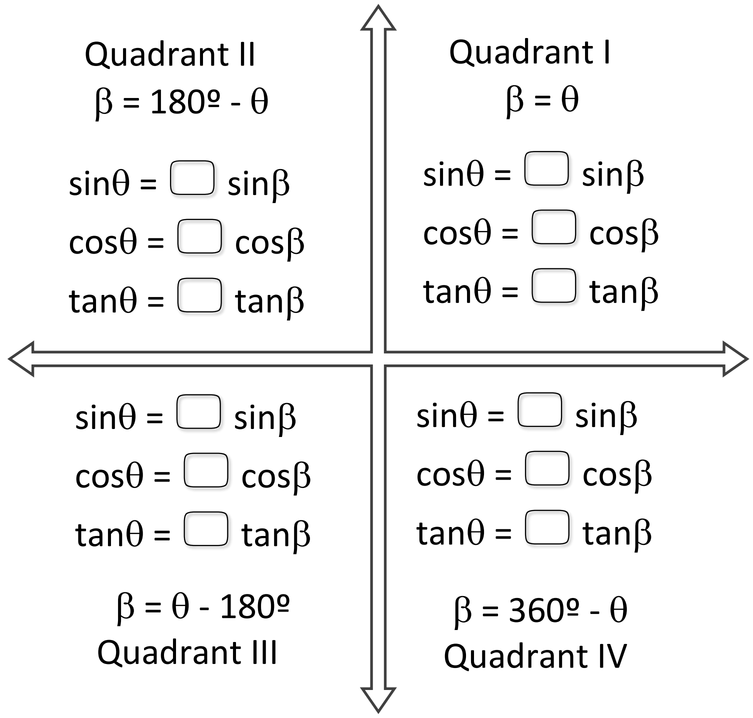
When θ is in Quadrant III.



When θ is in Quadrant IV.



The **CAST Rule** is an easy way to remember which trigonometric ratios are positive in each quadrant.



State the equivalent ratio for each of the following.

a) tan135°

b) cos240°

c) csc(-300°)

d) sec197°

e) cot(-55°)

f) sin281°

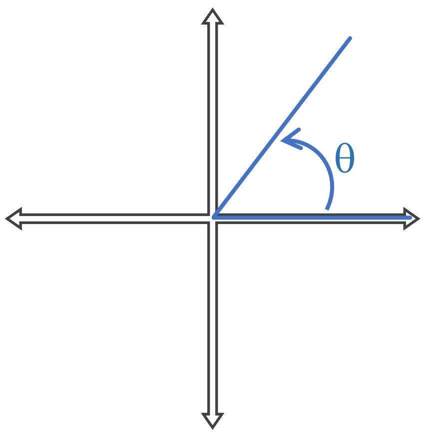
Solve for θ if $0° \leq \theta \leq 360°$.

a) $\cos\theta = \frac{5}{12}$

b) $\tan\theta = -\frac{11}{6}$

c) $\csc\theta = -\frac{7}{4}$

Exploring Trigonometric Ratios for Angles $\theta > 360°$



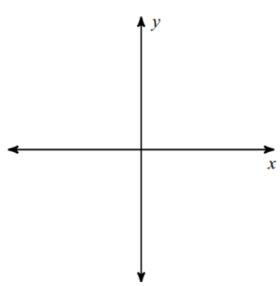
Angles that are one (or more) revolutions apart have a common ___________ ____. These angles are called ______________ ________.

_________________ _______ differ by a multiple of _____.

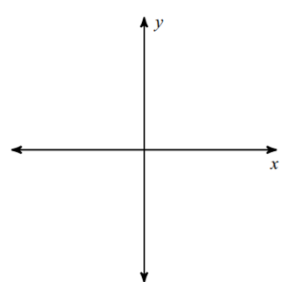
$$\alpha = \theta + 360°k, k \in Z$$

Sketch α and determine θ.

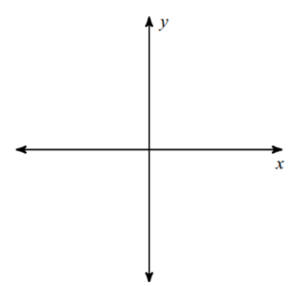
a) $\alpha = 435°$



b) $\alpha = 938°$



c) $\alpha = -527°$



State the equivalent ratio for the related acute angle for each of the following.

a) $\tan 755°$

b) $\sin(-600°)$

c) $\sec 556°$

Solve for θ if $0° \leq \theta \leq 720°$.

a) $\sin\theta = \frac{5}{12}$

b) $\cot\theta = -\frac{11}{6}$

c) $\sec\theta = -\frac{7}{4}$

Practice Questions

1. Sketch α and determine θ.

   a) $\alpha = 460°$
   b) $\alpha = 510°$
   c) $\alpha = 627°$
   d) $\alpha = 935°$
   e) $\alpha = -700°$
   f) $\alpha = -491°$
   g) $\alpha = -1080°$
   h) $\alpha = -845°$

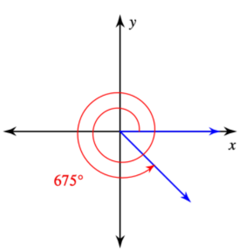
2. Find a positive and a negative coterminal angle for each given angle.

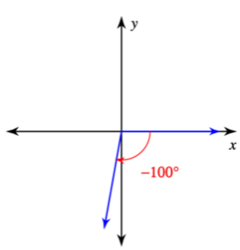
   a) $150°$
   b) $635°$
   c) $-330°$
   d) $-750°$

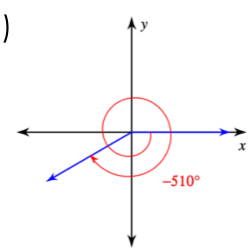
3. State if the given angles are coterminal.

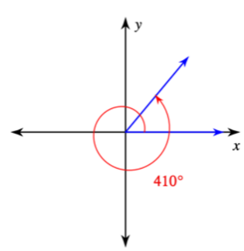
   a) $90°, 630°$
   b) $275°, -635°$
   c) $285°, 645°$
   d) $230°, -590°$

4. Find the related acute angle, β, for each below.

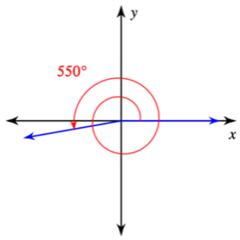
   a)


   b)


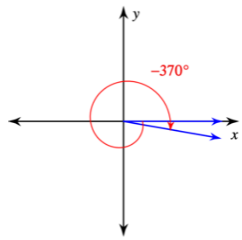
   c)


   d)


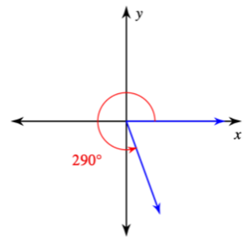
e)



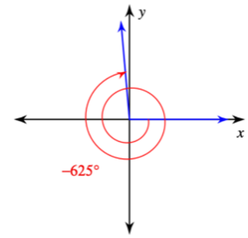
f)



g)



h)



5. State the equivalent ratio for the related acute angle for each of the following.

   a) $\sin 525°$ b) $\cot(-407°)$ c) $\cos 912°$

   d) $\sec(-710°)$ e) $\tan 1250°$ f) $\csc(-638°)$

6. Determine 4 coterminal angles (2 positive and 2 negative) that share a common terminal arm containing the point (-4, 11).

7. Determine the approximate measure of all angles that satisfy the following. Give answers to two decimal places. Use diagrams to show the possible answers.

   a) $\cos\theta = 0.42$ in the domain $0° \leq \theta \leq 720°$ b) $\tan\theta = -4.87$ in the domain $-360° \leq \theta \leq 360°$

   c) $\csc\theta = -1.5$ in the domain $-180° \leq \theta \leq 180°$ d) $\sin\theta = 0.65$ in the domain $0° \leq \theta \leq 1080°$

   e) $\cot\theta = \frac{17}{6}$ in the domain $-90° \leq \theta \leq 450°$ f) $\sec\theta = -\frac{15}{13}$ in the domain $-720° \leq \theta \leq 0°$

8. Given angle θ, where $0° \leq \theta \leq 360°$, solve for all possible values of θ to the nearest degree.

   a) $\cos 2\theta = 0.6420$ b) $\sin(\theta - 20°) = 0.2045$ c) $\tan(90° - 2\theta) = 1.6443$

Answers
1.a) 100° b) 150° c) 267° d) 215° e) 20° f) 229° g) 0° h) 235° 2. (Answers may vary.) a) -210 °, 510° b) 275°, -85° c) 30°, -690° d) -390°, 330° 3.a) No b) No c) Yes d) No 4.a) 45° b) 80° c) 30° d) 50° e) 10° f) 10° g) 70° h) 85° 5.a) $\sin 15°$ b) $-\cot 47°$ c) $-\cos 12°$ d) $\sec 10°$ e) $-\tan 10°$ f) $\csc 82°$ 6. -250°, -610°, 110°, 470° 7. a) 65.17°, 294.83°, 425.17°, 654.83° b) -258.40°, -78.40°, 101.60°, 281.60° c) -138.19°, -41.81° d) 40.54°, 139.46°, 400.54°, 499.46°, 760.54°, 859.46° e) 19.44°, 199.44°, 379.44° f) -569.93°, -510.07°, -209.93°, -150.07° 8. a) 25°, 155°, 205°, 335° b) 32°, 188° c) 16°, 106°, 196°, 286°

Special Triangles: Exact Trigonometric Ratios of Special Angles

The 45º-45º-90º Special Triangle

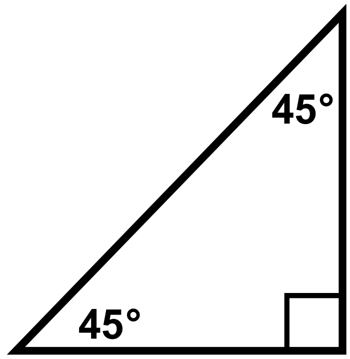
Special because we can know the ratios of its sides with simple

Since the angles are congruent, this triangle is classified as ___________, which means that the sides opposite the angles are then __________ as well.

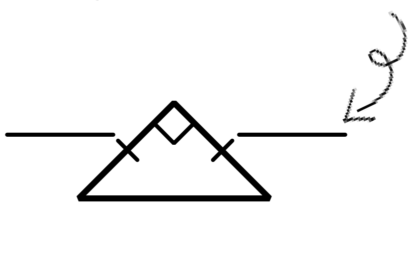
Ratio of sides

1 : 1 :

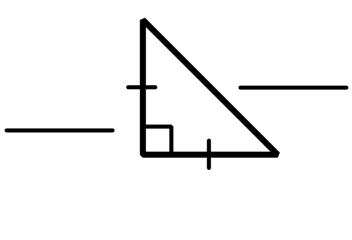
Use the Pythagorean Theorem to find the hypotenuse.



1. If the legs measured ______, then:



2. If the hypotenuse were ______, then:



Trigonometric Ratios:

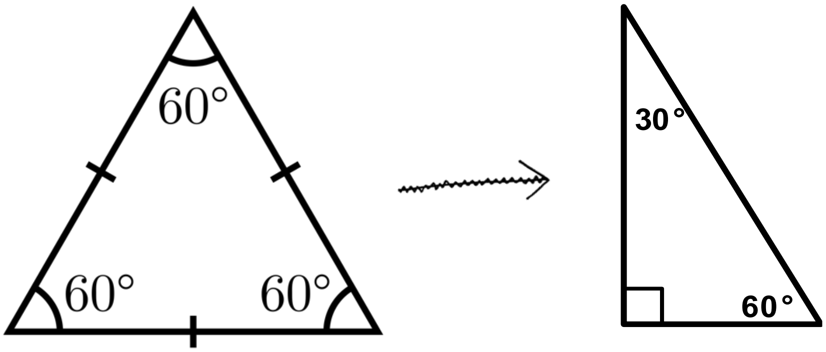
sin45º =

csc45º =

cos45º =

sec45º =

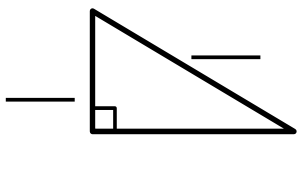
The 30º-60º-90º Special Triangle



Use the Pythagorean Theorem to find the missing side.

If the smallest side were ______,

then:



Ratio of sides

: : 1

1 : : 2

Trigonometric Ratios:

sin30º =

csc30º =

cos30º =

sec30º =

Trigonometric Ratios:

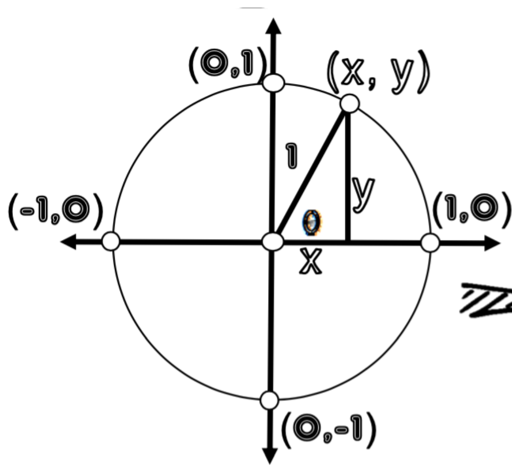
sin60º =

csc60º =

cos60º =

sec60º =

Introducing the Unit Circle (a circle on the cartesian plane with a radius of 1):



For an angle, θ, in standard position whose terminal ray passes through the point (x, y) on the unit circle:

$\sin(\theta)$ = the y-coordinate and

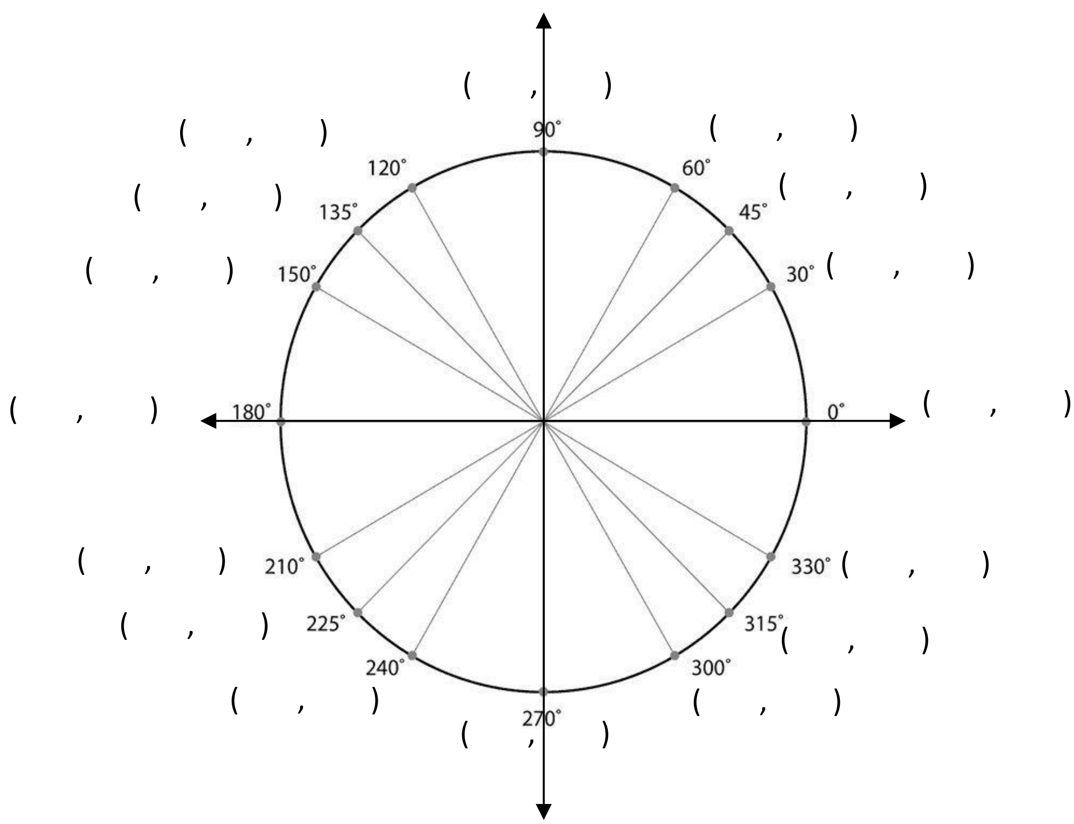
$\cos(\theta)$ the x -coordinate

WHY?

SOH CAH TOA

So, on the unit circle

$(\cos\theta, \sin\theta) = (\quad , \quad)$.



Ex 1: Find the exact values of each trigonometric ratio.

a) $\sin 225°$ b) $\cos 300°$ c) $\tan 570°$ d) $\sec 150^{\circ}$

Ex 2: Find the exact value of each of the following.

a) $\dfrac{3\tan 60^{\circ}}{\cos 30^{\circ}}$ b) $2\sin 135° + \frac{1}{2\cos(-225°)}$

Ex 3: If $0° \le \theta \le 360°$, find the possible measures of θ.

a) $\tan\theta = \sqrt{3}$ b) $\sin\theta = -\frac{1}{2}$ c) $\cos\theta = -\frac{\sqrt{2}}{2}$

Ex 4: A flagpole is anchored by two guy wires attached 8 m above the ground. One guy wire is anchored to the ground at a 45° angle. The other guy wire is anchored to the ground at a 60° angle. How long is each wire?

Special Triangles: Practice

1. Find the exact value of each trigonometric ratio.

a) $\sin 60°$ b) $\cos 240°$ c) $\cot 150°$ d) $\cos 225°$ e) $\csc 45°$

f) $\cos 315°$ g) $\sin 300°$ h) $\tan 120°$ i) $\sec 150°$ j) $\tan 225°$

k) $\sin 315°$ l) $\tan 210°$ m) $\csc 225°$ n) $\cot 150°$ o) $\sec 210°$

2. Evaluate each of the following. Give exact answers.

a) $\sin^2 30° + \cos^2 30°$ b) $\tan^2 60° + 2\tan^2 45°$ c) $(\sin 30°)(\cos 45°)(\tan 60°)$

d) $\frac{2}{\sin^2 45°} - \frac{3}{\cos^2 30°}$ e) $2\sin 45° + \frac{1}{2\cos 45°}$ f) $\sin 30° + \cos 30°$

g) $(\sin 60° + \cos 30°)(\sin 60° - \cos 30°)$ h) $(\tan^2 45°)(\sin 60°)(\tan 30°)(\tan^2 60°)$

3. If $0° \le \theta \le 360°$, find the possible measures of θ.

a) $\sin\theta = \frac{-\sqrt{3}}{2}$ b) $\cos\theta = \frac{1}{2}$ c) $\tan\theta = \sqrt{3}$ d) $\cos\theta = \frac{\sqrt{3}}{2}$

e) $\tan\theta = -1$ f) $\cos\theta = \frac{\sqrt{2}}{2}$ g) $\sin\theta = \frac{\sqrt{3}}{2}$ h) $\tan\theta = \frac{-\sqrt{3}}{3}$

4. Find the exact values of the six trigonometric ratios for each angle.

a) 390° b) 405° c) 690° d) 450°

e) -60° f) -225° g) -510° h) -540°

5. Find the possible measures for θ in the desired interval.

a) $\sin\theta = \frac{-\sqrt{2}}{2}, 0° \le \theta \le 720°$ b) $\sec\theta = 2, -180° \le \theta \le 180°$

c) $\tan\theta = 1, -540° \le \theta \le 360°$ d) $\csc\theta = \frac{-2\sqrt{3}}{3}, 0° \le \theta \le 720°$

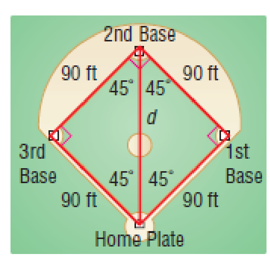
e) $\cos\theta = \frac{\sqrt{3}}{2}, -360° \le \theta \le 360°$ f) $\cot\theta = -\sqrt{3}, -180° \le \theta \le 540°$

6. Given angle θ, where $0° \le \theta \le 360°$, solve for θ to the nearest degree.

$$\sin(30° + 2\theta) = \frac{\sqrt{3}}{2}$$

7. Find the six trigonometric ratios of 0°, 180°, 270°, and 360° angles.

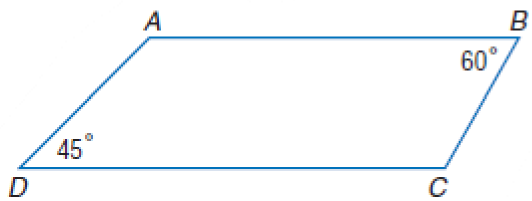
8. Find the distance from home plate to second base if the bases are 90 feet apart in exact values.



9. The diagonals of a rectangle are 12 cm long and intersect at an angle of 60°. Find the perimeter of the rectangle in exact values.

10. The boom crane can be moved so that its angle of inclination changes. In one location, close to some buildings and some overhead power cables, the minimum value of the angle of inclination is 30°, and the maximum value is 60°. If the boom is 10 m long, find the vertical displacement of the end of the boom when the angle of inclination increases from its minimum value to its maximum value. Express your answer in exact values.

11. Given figure ABCD, with $\overline{AB} \parallel \overline{DC}, m\angle B = 60°, m\angle D = 45°, BC = 8$, and $AB = 24$, find the perimeter in exact values.



Solutions

1. a) $\frac{\sqrt{3}}{2}$ b) $-\frac{1}{2}$ c) $\frac{-\sqrt{3}}{3}$ d) $\frac{-\sqrt{2}}{2}$ e) $\sqrt{2}$ f) $\frac{\sqrt{2}}{2}$ g) $\frac{-\sqrt{3}}{2}$ h) $-\sqrt{3}$
i) $\frac{-2\sqrt{3}}{3}$ j) 1 k) $\frac{-\sqrt{2}}{2}$ l) $\frac{\sqrt{3}}{3}$ m) $-\sqrt{2}$ n) $-\sqrt{3}$ o) $\frac{-2\sqrt{3}}{3}$

2. a) 1 b) 5 c) $\frac{\sqrt{6}}{4}$ d) 0 e) $\frac{3\sqrt{2}}{2}$ f) $\frac{1+\sqrt{3}}{2}$ g) 0 h) $\frac{3}{2}$

3. a) θ = 240° and θ = 300° b) θ = 60° and θ = 300° c) θ = 60° and θ = 240°
d) θ = 30° and θ = 330° e) θ = 135° and θ = 315° f) θ = 45° and θ = 315°
g) θ = 60° and θ = 120° h) θ = 150° and θ = 330°

4. a) $\sin 390° = \frac{1}{2}, \cos 390° = \frac{\sqrt{3}}{2}, \tan 390° = \frac{\sqrt{3}}{3}, \csc 390° = 2, \sec 390° = \frac{2\sqrt{3}}{3}, \cot 390° = \sqrt{3}$

b) $\sin 405° = \frac{\sqrt{2}}{2}, \cos 405° = \frac{\sqrt{2}}{2}, \tan 405° = 1, \csc 405° = \sqrt{2}, \sec 405° = \sqrt{2}, \cot 405° = 1$

c) $\sin 690° = -\frac{1}{2}, \cos 690° = \frac{\sqrt{3}}{2}, \tan 690° = -\frac{\sqrt{3}}{3}, \csc 390° = -2, \sec 390° = \frac{2\sqrt{3}}{3}, \cot 390° = -\sqrt{3}$

d) $\sin 450° = 1, \cos 450° = 0, \tan 450° = und, \csc 450° = 1, \sec 390° = und, \cot 450° = 0$

e) $\sin(-60°) = -\frac{\sqrt{3}}{2}, \cos(-60°) = \frac{1}{2}, \tan(-60°) = -\sqrt{3}, \csc(-60°) = -\frac{2\sqrt{3}}{3}, \sec(-60°) = 2, \cot(-60°) = -\frac{\sqrt{3}}{3}$

f) $\sin(-225°) = \frac{\sqrt{2}}{2}, \cos(-225°) = -\frac{\sqrt{2}}{2}, \tan(-225°) = -1, \csc(-225°) = \sqrt{2}, \sec(-225°) = -\sqrt{2}, \cot(-225°) = -1$

g) $\sin(-510°) = \frac{1}{2}, \cos(-510°) = -\frac{\sqrt{3}}{2}, \tan(-510°) = -\frac{\sqrt{3}}{3}, \csc(-510°) = 2, \sec(-60°) = -\frac{2\sqrt{3}}{3}, \cot(-510°) = -\sqrt{3}$

h) $\sin(-540°) = 0, \cos(-540°) = -1, \tan(-540°) = 0, \csc(-540°) = und, \sec(-60°) = -1, \cot(-540°) = und$

5. a) 225°, 315°, 585°, 675° b) -60°, 60° c) -495°, -315°, -225°, 45°, 225° d) 240°, 300°, 600°, 660°
e) -330°, -30°, 30°, 330° f) -150°, 150°, 210°, 510°

6. 15°, 45°, 195°, 225°

7. $\sin 0° = 0, \cos 0° = 1, \tan 0° = 0, \csc 0° = und, \sec 0° = 1, \cot 0° = und;$ $\sin 90° = 1, \cos 0° = 0, \tan 0° = und, \csc 0° = 1,$ $\sec 90° = und, \cot 90° = 0;$ $\sin 180° = 0, \cos 180° = -1, \tan 180° = 0, \csc 180° = und, \sec 180° = -1, \cot 180° = und;$
$\sin 270° = -1, \cos 270° = 0, \tan 270° = und, \csc 270° = -1, \sec 270° = und, \cot 270° = 0;$ $\sin 360° = 0, \cos 360° = 1,$ $\tan 360° = 0, \csc 360° = und, \sec 360° =, \cot 360° = und$

8. $90\sqrt{2}$ ft

9. $12 + 12\sqrt{3}$ cm

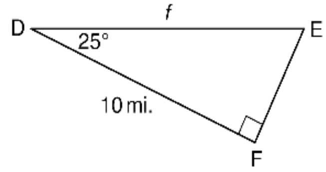
10. $5\sqrt{3} - 5$ m

11. $52 + 4\sqrt{3} + 4\sqrt{6}$ units

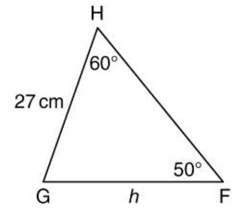
Relax… Review!

1. Find the indicated side or angle to the nearest tenth .

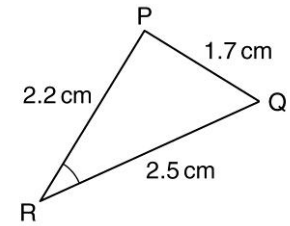
a)



b)



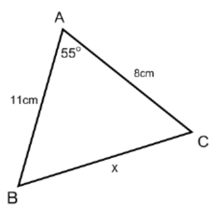
c)



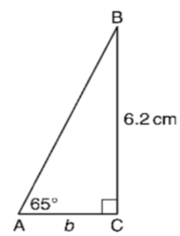
2. Two angles of a triangle are 57° and 83° and the shortest side is 14 m. Sketch the triangle and briefly explain how you know where the shortest side should be placed.

3. For the following triangles, clearly state and explain which method you would use to solve for x. (SOH CAH TOA, the Sine Law, or the Cosine Law) If your answer is SOH CAH TOA, explain which ratio you would use and why.

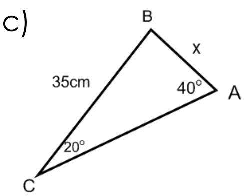
a)



b)



c)



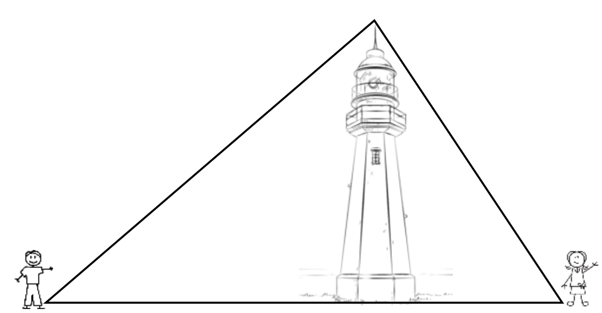
4. A wheelchair ramp is 3.2 m long. It rises from the ground to a porch that is 0.35 m above the ground. Determine the angle of elevation for the ramp. Include a diagram.

5. A bus station is at the intersection of Main St. and King St. The angle between the streets is 67°. Two buses leave the bus station at the same time. After 15 min, one bus has travelled 2.5 km along Main St. The other bus has travelled 5.0 km along King St. How far apart are the buses? Include a diagram.

6. Andrew is flying a kite in a flat, open field. He lets out 15 m of string. His friend Roy is also playing in the field, 26 m from Andrew. The angle of elevation of the kite from Roy's position is 35°. Determine the angle of elevation of the kite from Andrew's position. Include a diagram.

7. The bases in a baseball diagram are 90 ft apart. A player picks up a ground ball 20 ft from third base along the line from second base to third base and throws it to the first baseman. Determine the angle that is formed between home plate, the player's present position and the first baseman.

8. Bogdan and Chloe are standing 70 ft. apart on opposite sides of the base of a lighthouse. The angle of elevation from each person to the top of the lighthouse is 37° from Bogdan and 56° from Chloe. How tall is the lighthouse?



9. ∠K is in the third quadrant with a secant of –6.
   a) Sketch ∠K and find the exact expressions for the other five trigonometric ratios for ∠K.
   b) Determine ∠K.
   c) Determine one positive and one negative co-terminal angle for ∠K.

10. Find the possible measures for θ, to the nearest tenth, in the desired interval.

a) $\sin\theta = \frac{2}{3}, 0° \le \theta \le 360°$

b) $\sec\theta = -1.62, -180° \le \theta \le 180°$

c) $\cot\theta = -0.27, 0° \le \theta \le 720°$

d) $\csc\theta = -\frac{5}{4}, 0° \le \theta \le 1080°$

e) $\cos\theta = \frac{\sqrt{3}}{2}, 0° \le \theta \le 360°$

f) $\tan\theta = -\sqrt{3}, -180° \le \theta \le 540°$

11. Kim said she found three angles for which $\tan\theta = -1.249$. Is that possible if 0° ≤ θ ≤ 360°? Explain.

12. Determine the exact value of:

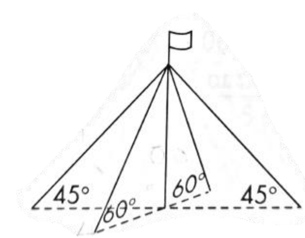
$$(\sin^2 240°)(\sec 135°) + \tan(-30°)(\csc 330°) - (\cos 405°)(\cot 120°)$$

13. Given angle θ, where $0° \le \theta \le 360°$, solve for θ to the nearest degree.

$$\sec(30° + 2\theta) = 1.5378$$

14. After heavy winds damaged a farmhouse, workers placed a 6-m brace against its side at an angle of 30°. Then, at the same spot on the ground, they placed a second, longer brace to make a 60° with the side of the house. How long is the longer brace in exact values?

15. A flagpole is anchored by two points of guy wires attached 8 m above the ground. One pair is anchored to the ground at 45° angles. The other pair is anchored to the ground at 60° angles. Which pair of guy wires has the greater total length and by how much, in exact values.



Answers

1. a) 11.0 mi b) 30.5 cm c) 41.8° 2. The shortest side will be across from the third unknown angle. 3. a) Cosine law b) Tangent ratio c) Sine law 4. 6.3° 5. 4.6 km 6. 61.2° 7. 50.4° 8. 21.7 ft 9. $\sin K = -\frac{\sqrt{35}}{6}, \cos K = -\frac{1}{6}, \tan K = \sqrt{35}$, $\csc K = -\frac{6\sqrt{35}}{35}, \cot K = \frac{\sqrt{35}}{35}$ b) 260° c) 620° and -100° 10. a) 41.8°, 138.2° b) -128.1°, 128.1° c) 105.1°, 285.1°, 465.1°, 645.1° d) 233.1°, 306.9°, 593.1°, 666.9°, 953.1°, 1026.9° e) 30°, 330° f) -60°, 120°, 300°, 480° 11. It is not possible. 12. $\frac{-9\sqrt{2}+8\sqrt{3}+2\sqrt{6}}{12}$ 13. 9.5°, 140.5°, 189.5°, 320.5° 14. $6\sqrt{3}$ m 15. The 45° guy wires by $\frac{48\sqrt{2}-32\sqrt{3}}{3}$ m.